

Batch vs Stochastic vs Mini-Batch Gradient Descent

All three variants minimize the loss by repeating $\text{params} = \text{params} - \text{learning_rate} * \text{params_grad}$. They differ in how much data is used to compute the gradient before each update.

1. Batch GD (Vanilla GD)

Uses the **entire dataset** for one update.

```
for i in range(nb_epochs):
    params_grad = evaluate_gradient(loss_function, data, params)
    params = params - learning_rate * params_grad
```

- **Updates per epoch:** 1
- **Pros:** Stable, smooth convergence; exact gradient.
- **Cons:** Slow on large datasets, memory heavy, can get stuck in local minima or saddle points.

2. Stochastic GD (SGD)

Updates the parameters after **each single example**.

```
for i in range(nb_epochs):
    np.random.shuffle(data)
    for example in data:
        params_grad = evaluate_gradient(loss_function, example, params)
        params = params - learning_rate * params_grad
```

- **Updates per epoch:** N (number of examples)
- **Pros:** Fast, frequent updates; low memory; noise can help escape local minima.
- **Cons:** Loss fluctuates, may never settle exactly at the minimum, poor use of vectorization.

3. Mini-Batch GD

The **best of both worlds**: updates on small batches (e.g. `batch_size=50`).

```
for i in range(nb_epochs):
    np.random.shuffle(data)
    for batch in get_batches(data, batch_size=50):
        params_grad = evaluate_gradient(loss_function, batch, params)
        params = params - learning_rate * params_grad
```

- **Updates per epoch:** N / batch_size
- **Pros:** More stable than SGD, faster than Batch GD, uses GPU/vectorized ops, memory efficient.
- **Cons:** Adds a hyperparameter (batch size) to tune; still some noise.

Comparison

	Batch GD	SGD	Mini-Batch GD
Data per update	Entire dataset	1 example	Small batch (32, 64, 128)
Updates per epoch	1	N	N / batch_size
Speed per update	Slow	Very fast	Fast
Convergence	Smooth, stable	Very noisy	Moderately smooth
Memory use	High	Low	Medium
Vectorization	Yes	No	Yes
Practical use	Small datasets	Online learning	Default in deep learning

Key Takeaways

- Batch GD is one end (all data), SGD is the other (one sample), Mini-Batch sits in between.
- Shuffle the data every epoch in SGD and Mini-Batch so updates aren't biased by data order.
- Common batch sizes are powers of 2 (32, 64, 128, 256), which map well to GPU memory.
- In deep learning libraries, "SGD" usually means mini-batch GD.